

Preliminary Proposal:

Predicting Cardiovascular Disease Using Clinical Features: A Machine Learning Approach

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Background and Question

In the context of clinical risk, can machine learning models accurately predict the presence of cardiovascular disease, and how should model performance be evaluated given the trade-off between sensitivity and specificity?

Cardiovascular disease (CVD) is the leading cause of death for both men and women worldwide, and machine learning has become a popular tool for trying to predict it. A lot of that work exists already. The problem is that most of it stops at accuracy and AUC scores without asking a more important question: what do the errors actually mean for a patient? A model that looks good on paper can still miss 40% of true cases, and in a clinical setting, that is a real consequence. This project is built around that concern, treating model evaluation as a clinical risk question rather than just a statistical one. The focus will not be on whether a model can predict CVD, but on whether it can be evaluated and validated in a way that makes it valuable for real-world clinical decision support. “A Comprehensive Review of Machine Learning For Heart Disease Prediction” supports this by finding that model selection should be based on performance metrics crucial for clinical diagnostic accuracy rather than overall accuracy alone (Kumar et al.,2025).

Platforms that detect coronary artery disease lesions using AI-driven diagnostic tools are becoming standard in clinical imaging environments. The ability to thoroughly assess model performance against clinical risk thresholds is a critical skill. Validation teams must ask themselves not only if a model is accurate, but also if the model error profile can be clinically trusted and safe for patient outcomes. This project will simulate the validation process to make it both academically rigorous and professionally meaningful.

Cardiovascular disease prediction using the UCI Heart Disease dataset is well used in machine learning publications. The project is not novel in that sense. The project will aim to reframe and improve the approach from a clinical validation standpoint and apply it to model evaluation. From an academic project level perspective, model evaluation is treated as a report of sensitivity and specificity metrics, but does not address those metrics with clinical design decisions and their real-world consequences. This perspective is standard in clinical AI development.

The primary stakeholder identified is AI/ML developers and clinical validation teams working in medical imaging and diagnostic AI. They are the ones responsible for assessing whether a model is safe for deployment in a clinical workflow. Secondary stakeholders include cardiologists and clinical informaticists who use and interpret these tools for patient point of care.

Hypothesis and Prediction

Optimizing a machine learning model’s sensitivity will reduce false negatives but increase false positives. In cardiovascular disease prediction this trade-off carries measurable clinical complications. A model cannot be evaluated as clinically appropriate without also analyzing what its error profile means in a real-world care setting.

A tuned ensemble model (Random Forest or XGBoost) optimized for sensitivity will outperform a baseline Logistic Regression model on recall and ROC-AUC. However, the most clinically appropriate model will not necessarily be the highest performing model based on accuracy, it will be the model whose sensitivity (true positive)/specificity (true negative) balance best aligns with defined clinical risk thresholds. Sensitivity will be prioritized as the primary performance target, as false negatives could have dire patient outcomes, more so than false positives. Sensitivity and specificity threshold goals are $\geq 85\text{-}90\%$ and $\geq 75\text{-}80\%$, respectively. These thresholds are chosen based on literature findings in a 2024 clinical feature analysis for CVD risk screening using XGBoost. A sensitivity of 89.1% and specificity of 85.39% were achieved and selected using the Matthews Correlation Coefficient, while keeping false negative rate clinically acceptable (Vyshnya et al, 2024).

Additionally, foundational clinical methods literature establishes that for cardiovascular diagnostic screening, a sensitivity of 75% and specificity of 80% represent a reasonable clinical benchmark, and that these metrics must always be interpreted together with disease prevalence and clinical context rather than isolation (Simon and Boring, 1990).

These thresholds reflect the clinical reality that a missed cardiovascular diagnosis carries greater risk than an unnecessary follow-up.

Dataset and Analysis:

Dataset

I will be using the UCI Heart Disease dataset, which can be found here:

<https://github.com/sharmaroshan/Heart-UCI-Dataset/blob/master/heart.csv>.

The dataset includes 303 patient records with 13 clinical features and a binary target variable for disease presence. It was originally collected across four institutions and is widely used and cited in cardiovascular ML research. The downside to this dataset is that it is older and limited in scope, so results will be treated as proof-of-concept rather than a deployable tool.

Response / Outcome Variable

Target (binary): 0 = No cardiovascular disease, 1 = Cardiovascular disease present.

Predictor Variables

All 13 clinical features will be used as candidate predictors: age, sex, chest pain type, resting blood pressure, serum cholesterol, fasting blood sugar, resting ECG results, maximum heart rate, exercise-induced angina, ST depression, ST slope, number of major vessels, and thalassemia type. Feature importance will be assessed during preprocessing.

Tentative Analysis Plan:

This project will include both supervised and unsupervised learning components.

PCA will be used for unsupervised to assess the variance structure of the dataset and reduce dimensionality before supervised modeling. This component of the plan may be refined as the project develops.

Three classification models will be built and compared for supervised learning. Logistic Regression as a baseline, and Random Forest and XGBoost as tunable ensemble methods. Both will go through cross-validation and hyperparameter tuning.

Evaluation:

Models will be evaluated on accuracy, precision, recall, specificity, F1-score, and ROC-AUC. A threshold analysis will also be run to see how shifting the classification cutoff affects the sensitivity/specificity tradeoff and what that means from a clinical standpoint.

Potential Pitfalls:

- Small sample size ($n=303$): This limits how well results will generalize and will be noted as a key limitation.
- This dataset is known to have a class imbalance. If one outcome is more dominant than the other than accuracy will be a poor metric. Sensitivity, specificity, and ROC-AUC will be used instead.
- Dataset age: The UCI dataset is an older dataset (1988) and may not reflect current patient populations. Findings will be scoped accordingly.
- Ensemble models can overfit on small datasets. Cross-validation and tuning will be used throughout.
- Reducing features to principal components makes individual variables harder to interpret clinically. This tradeoff will be addressed in the results.

How Will I Know if the Question Is Answered?

I will compare all three models once they have been trained and evaluated. I will compare their performance across clinical metrics and make a recommendation on which best fits a clinical decision support setting. The answer will be framed around what the results actually mean for a patient population, not just a report of the numbers.

How Will I Know if the Hypothesis Is Supported?

The hypothesis will be supported if there is a clear trade-off between sensitivity and specificity across models, and if tuning for sensitivity shifts the false negative/positive balance in a meaningful way. If that trade-off does not show up, that can still be useful and worth discussing.

Technical Details:

Programming Language: Python (Google Colab)

Additional Resources: scikit-learn, pandas, NumPy, matplotlib, seaborn, XGBoost

GitHub Repository: https://github.com/mnooney/CVD_Capstone

References

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- Simon D, Boring JR III. Sensitivity, Specificity, and Predictive Value. In: Walker HK, Hall WD, Hurst JW, editors. *Clinical Methods: The History, Physical, and Laboratory Examinations*. 3rd edition. Boston: Butterworths; 1990. Chapter 6. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK383/>